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Ground Truth Estimator

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User Guide

First-Principles Estimating, and How to Use This Software

This guide has two parts. Part One explains first-principles estimating itself — the considerations a proper estimator works through before a number ever gets typed in. Part Two walks through Ground Truth Estimator step by step, in the order you'll actually use it, from setting up your workspace to sending out a finished estimate.

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CONTENTS

What's in this guide

Part One — First-Principles Estimating

Part Two — Using Ground Truth Estimator

PART ONE

First-Principles Estimating

“First-principles” estimating means building a price up from its real components — labour hours, plant hours, and material quantities — rather than pulling a single per-metre or per-square-metre figure from memory or from a past job that may not be comparable. It's slower than a gut-feel number, but it produces a price you can actually defend, line by line, if a client, a quantity surveyor, or your own finance team asks “where did this number come from?”

> *“A single lump-sum rate is a guess wearing a suit. A build-up is a guess you can cross-examine.”*

The ten considerations behind every sound estimate

These apply whatever software you use — they're the discipline itself, not a feature list. Ground Truth Estimator is built to walk you through them in this order.

1. Understand the scope and the site

Before any number is typed, read the drawings, specification, and site conditions. Access, ground conditions, staging area, existing services, and site constraints all change what labour and plant an item actually needs — the same trench costs differently in open paddock versus a live urban road.

2. Break the work into logical categories

Group the project into work packages that make sense for how it will actually be built and managed — typically earthworks and site works, roads and pavements, drainage and pipelines, and structures. This keeps the estimate organised and makes it possible to compare category totals against past jobs.

3. Quantify each item properly (quantity take-off)

Measure real quantities off the drawings and specification — lengths, areas, volumes, counts. An estimate is only as good as its quantities; a first-principles rate applied to a wrong quantity still produces a wrong price.

4. Build the rate up from labour, plant, and material

For every line item, work out what labour it actually takes (whose time, how many hours), what plant it needs (machine and hire hours), and what material goes into it (quantity and unit cost) — then add those three together to get the unit rate. This is the core of “first principles.”

5. Keep units of measure consistent

A rate quoted per linear metre is meaningless multiplied against a quantity measured in square metres. Confirm every item's unit matches how it was actually quantified, and be explicit when converting between metric and imperial so nothing gets lost in translation.

6. Assess risk as its own exercise — not a blanket percentage

Rather than adding a flat “10% for contingency” across the board, list out the specific things that could go wrong — weather delays, unknown ground conditions, market price movement, programme slippage, safety incidents — and estimate a probability and a cost impact for each. The sum of those gives a contingency figure you can actually explain, item by item, rather than a number picked to feel safe.

7. Weigh location- and season-specific risk

Where and when the work happens matters. A project programmed through a wet season or a cyclone-prone period carries real, quantifiable weather risk that a generic contingency percentage won't capture.

8. Apply markups deliberately, and keep tax separate

Preliminaries / Indirect Job Costs (site establishment, supervision, temporary works), overhead (your business's running costs), and margin (profit) are three different things and should be considered separately, not lumped into one "mark-up." Sales tax (GST, VAT, or your local equivalent) is not a cost to the business at all — it should be shown separately, not folded into the price you're trying to win the job on.

Where you have enough information to do it properly, apply this same discipline to preliminaries itself: build it up from the real components — site supervision, temporary services, site facilities, insurances, bonds — rather than estimating a flat percentage of direct cost. See Step 6 in Part Two.

9. Sanity-check the total

Once the build-up is complete, step back and compare the total — and the category breakdown — against past projects, industry benchmarks, or a rough top-down check. A first-principles estimate that comes out wildly different from experience is a signal to go back and check assumptions, not necessarily proof the number is right.

10. Document assumptions and keep rates current

Rates go stale. Labour agreements change, plant hire costs move, material prices fluctuate. A rate library is only trustworthy if it's actively maintained against real supplier quotes and current agreements — treat every stored rate as a placeholder until you've checked it recently.

PART TWO

Using Ground Truth Estimator, Step by Step

The software mirrors the discipline in Part One. Used in this order, it walks you through the same considerations a first-principles estimator works through by hand — just faster, and with every figure stored, reusable, and traceable.

1 Set up your workspace

When you create an account, a workspace is created for your company. Everything you build — rates, projects, risk registers — belongs to that workspace and is never visible to anyone outside your organisation.

Before doing anything else, go to Settings and set your workspace's currency and preferred unit of measure (metric or imperial). Every dollar figure in the app formats to that currency, and any rate or quantity entered in the other unit system will show a small converted equivalent alongside it — without changing what you actually typed.

Where: Left rail → Settings

2 Build your Rate Library first

Before starting a project, populate the Rate Library with your own real labour, plant, and material rates — not the indicative placeholders the app starts with. This is the foundation everything else is built from, so it's worth getting right before you start pricing actual work.

Add rows under Labour, Plant & Equipment, and Material. Each rate needs a description, a unit, and a cost per that unit. If you're unsure of a current market rate, the “Search the web” button next to each row opens a web search for that item so you can check current supplier or hire pricing — the search results are for your own reference; nothing is applied automatically.

Where: Left rail → Rate Library

3 Start a project

From Projects, create a new project — either blank, or from the example project if you want to see a fully worked estimate first. Give it a name, client, location, and prepared-by details; these appear on exports.

A new project comes pre-loaded with four standard categories (Earthworks & Site Works, Roads & Pavements, Drainage & Pipelines, Structures), which you can rename, reorder, or add to as the project requires.

Where: Left rail → Projects → New Project

4

Build the estimate

Inside a project, the Estimate tab is where the first-principles build-up actually happens. Add line items under each category, give each a description, unit, and quantity, then build up its rate from your Rate Library — pulling in the specific labour, plant, and material components that make up that item, each with its own per-unit consumption.

The line item's rate is the sum of those components — exactly the discipline described in Part One. If a component's unit is in the other measurement system from your workspace default, a small converted figure appears alongside it so you can sanity-check without doing the maths by hand.

Where: Project → Estimate tab

5

Work through Risk & Location

Move to the Risk & Location tab and build the project's risk register as its own exercise, separate from the base estimate — add each identified risk with a description, a probability (as a %), and a cost impact if it occurs. The tab totals these into a risk allowance you can see and defend line by line, rather than a single guessed contingency percentage.

Enter the site location once and use it for four live lookups. The **weather risk lookup** pulls historical climate data — average rainfall, rainy days, and cyclone-zone exposure. The **geotechnical / soil risk lookup** checks free government soil survey data (USDA SSURGO in the United States, CSIRO's national Australian Soil Classification in Australia) and can flag reactive clay, poor drainage, or a shallow water table. The **flood risk lookup** checks FEMA's National Flood Hazard Layer for US sites, and the **seismic risk lookup** checks USGS Design Maps (US) and Geoscience Australia's National Seismic Hazard Assessment (AU) — both can flag an elevated hazard level worth a specific allowance.

Each lookup that finds something relevant can suggest a pre-filled risk item with a probability, ready to add straight to the register or adjust first. Outside the regions a lookup covers, it will say so plainly rather than guess — add a risk row manually instead. All four are a desktop-level early warning, not a substitute for a real site investigation, the actual FEMA FIRM panel, or a project-specific seismic assessment.

Where: Project → Risk & Location tab

6 Review the Summary and export

The Summary tab rolls everything up: direct costs by category, the risk allowance, and your markups — preliminaries / indirect job costs, contingency, overhead, and margin — applied on top. Sales tax (GST / VAT / Sales tax, depending on your workspace) is shown as a separate line, not folded into the price itself, and there's a separate line for client-side administrative cost where relevant.

Preliminaries / Indirect Job Costs can be priced two ways. Simple % (the default) applies a flat percentage of direct cost — quick, and reasonable for an early, rough-order estimate. Itemised build-up applies the same first-principles discipline from Part One to preliminaries itself: instead of a guessed percentage, list the real components — site supervision, temporary services, site facilities, insurances, bonds and guarantees, mobilisation — each priced as either a one-off Fixed cost or a $/week Time-related cost. Time-related items are multiplied automatically by the project duration you enter, so extending the programme extends every time-related item with it, rather than needing to be re-typed. Switching between the two modes doesn't lose anything — your itemised list stays saved even while Simple % is selected.

Two **Quick add** panels sit above the itemised list so common items don't need to be typed from scratch: one for on-site overhead and supervisory roles — Project Director, Project Manager, Site Manager / Construction Manager, Site Engineer, Quality Manager, Environmental Manager, Safety Officer, and others, plus a Custom role option — and one for insurances, bonds & guarantees, permits, and mobilisation-type pay items. Pick an item, set its rate, and it's added to the list ready to fine-tune or re-categorise.

Adjust markup percentages here and watch the total update live. When the estimate is ready, export it to CSV for use in a proposal, tender, or your own reporting.

Where: Project → Summary tab

7 Keep it current

A first-principles estimate is only as good as the rates behind it. Revisit the Rate Library periodically against real supplier quotes, current labour agreements, and plant hire pricing — the “Search the web” button is there specifically to make that check quick.

If you ever need to reset your password, use the “Forgot password?” link on the login screen; if you try to create a second account with an email that's already registered, the app will tell you directly rather than leaving you wondering why no email arrived.

QUICK REFERENCE

Where do I…?

I want to…	Go to
Set my workspace's currency or unit system	Settings
Add or edit a labour, plant, or material rate	Rate Library
Check a current market rate for something	Rate Library → “Search the web” button
Start a new project	Projects → New Project
Add a category or line item to an estimate	Project → Estimate tab
Build up a line item's rate from labour/plant/material	Project → Estimate tab → line item
Add a risk, or run the weather/geotech/flood/seismic lookup	Project → Risk & Location tab
Adjust preliminaries / indirect job costs, contingency, overhead, or margin	Project → Summary tab
Switch preliminaries / indirect job costs to an itemised build-up	Project → Summary tab → Preliminaries
Quick-add a common site-staff role	Project → Summary tab → Preliminaries → Quick add
Quick-add an insurance, bond, or other pay item	Project → Summary tab → Preliminaries → Quick add
Export the estimate	Project → Summary tab → Export CSV
Reset a forgotten password	Login screen → Forgot password?
Re-read this guide	Left rail → Help

Your account

Your workspace and everything in it — rates, projects, risk registers — is private to your organisation. Signing in with an email that already has an account will tell you so directly rather than silently failing; use “Forgot password?” on the login screen at any time to regain access without contacting support.

This guide covers the app as it stands today and will be updated as new features are added. If something here doesn't match what you see on screen, the in-app Help page always reflects the current version.